

NEEHAR SAWANT

neeharsawant20@gmail.com • 832-701-9995

EDUCATION

University of Illinois at Urbana-Champaign Bachelor of Science in Electrical Engineering
Champaign, IL Minor in Computer Science
GPA: 3.77/4.00
James Scholar Engineering Honors
Dean's List Honor

Related Coursework: Operating Systems, Data Structures, Algorithms, Databases, Artificial Intelligence, Digital Systems, Semiconductors, Analog/Digital Signal Processing, Fields and Waves, Discrete Math

EXPERIENCE

Texas Instruments – *Software Engineer*, Dallas, TX August 2023 – Present

- Providing technical support for customer issues integrating RTOS SDK on automotive SoC; issues tracked by Jira
- Contributing to driver C-code for DMA, GPIO, UART, and I2C peripherals running on ARM Cortex-R5
- Implementing automotive functional safety mechanisms and requirements in driver software for ASIL-D rating
- Building ADAS demos to demonstrate vast capabilities of automotive SoC and exhibited at CES and Electronica
- Involved in customer debug calls to provide real-time solutions for complex ADAS applications

Apple – *Test Engineer Intern*, Sunnyvale, CA May 2022 – August 2022

- Developed automation tool in Python with GUI to compare test coverage between stations for improved efficiency
- Designed flex cable to test iPhone component while maintaining test results, module power, and signal integrity
- Worked with contract manufacturers to analyze display component testing builds for maximum yield and retest rates
- Performed data analysis on Face ID module using JMP to prevent assembly errors and ensure accurate functionality

Honeywell Aerospace – *Test Engineer Intern*, Kansas City, MO June 2021 – August 2021

- Used Python and MATLAB to pipeline dataset creation from video to train machine learning models for error prevention
- Developed unit and integrated tests in VB.NET for tester calibration software to ensure precise functionality
- Assisted with troubleshooting and calibration for radio frequency (RF) testing of radar and nuclear components
- Obtained Q-level security clearance from DOE, equivalent to Top Secret, to access innovative technology in labs

John Deere – *Software Engineer Intern*, Virtual June 2020 – August 2020

- Applied Python and AWS to create image augmentation utilities to increase data scientists' productivity and consistency
- Manipulated image and updated JSON labels to build utilities such as resize, contrast, blur, etc.
- Structured utilities into downloadable package with Jenkins job for universal convenience and normalized use
- Collaborated with Computer Vision and Machine Learning team in a virtual office environment

University of Illinois – *Undergraduate Research Assistant*, Champaign, IL August 2020 – May 2023

- Used React and JavaScript to implement Logic Gate Simulator app for students to virtually learn about digital logic
- Created Digital Multimeter app for students to study resistance networks and calculate voltage/current on breadboard
- Contributed to teaching curriculum to expose underrepresented students to STEM while impacting the community

PROJECTS

Expense Tracking Application May 2025 – Present

- Developed application using MERN stack enabling users to seamlessly track and manage transactions in real time
- Integrated Plaid APIs to securely connect user's bank accounts and automate transaction retrieval and categorization

Linux Based Operating System Design August 2021 – December 2021

- Designed and coded kernel for a Linux-like operating system in C and x86 Assembly
- Included support for context switching, scheduling, file system, segmented memory protection, keyboard driver, etc.

ADDITIONAL INFORMATION

Languages: Python, C++, C, SQL, JavaScript, HTML/CSS, MATLAB

Frameworks/Databases: MySQL, MongoDB, React, Flask, .NET, Neo4j

Certifications: Bootlin Linux Device Drivers, AWS Cloud Practitioner

Work Eligibility: Eligible to work in the U.S. with no restrictions